25901606 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBCH100 - ESSENTIALS OF BIOCHEMISTRY : VITAMINS, HORMONES, ENZYMES AND NEUROTRANSMITTERS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which coenzyme is derived from Pantothenic acid (Vitamin B5)? [U] / [CO1]

   a). Biotin b). Coenzyme Q 10

   c). Coenzyme A (CoA) d). Tetrahydrofolate

2. Find the reaction catalyzed by flavin adenine dinucleotide. [K] / [CO1]

   a). Redox reaction

   b). Decarboxylation

   c). Transamination

   d). Carboxylation

3. Which of the following neurotransmitter is an example of an inhibitory neurotransmitter? [K] / [CO4]

   a). GABA

   b). Substance P

   c). Aspartate

   d). Glutamate

4. Select the main function of oxidoreductase enzymes from the given options. [E] / [CO5]

   a). Hydrolysis of complex molecules b). Transfer of functional groups

   c). Oxidation-reduction reactions d). Isomerization of molecules

5. What is the significance of enzyme specificity in enzyme regulation? [E] / [CO5]

   a). It allows for the regulation of enzyme activity

   b). It enables the catalysis of multiple reactions

   c). It increases the efficiency of enzyme catalysis

   d). It is essential for the proper functioning of metabolic pathways

[1x5 = 5]

Part B

Fill In The Blank Questions

Answer all questions

Each question carries **2** marks

6. What is the daily requirement of vitamin B12 during pregnancy and lactation? [K] / [CO1]

7. .....is the form of vitamin D present in plants [K] / [CO1]

8. _____ is known as the father of endocrinology, and he discovered _____ disease. [K] / [CO2]

9. Antidiuretic hormone is related to the disorder_____ [U] / [CO3]

10. The temperature at which maximum enzymatic activity occurs is called------- [K] / [CO5]

[2x5 = 10]

Part C

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11.	What is xerophthalmia, and how is it related to Vitamin A deficiency?	[K] / [CO1]
12.	How is the secretion of oxytocin influencing social behavior and bonding?	[U] / [CO4]
13.	Describe the role of G-protein-coupled receptors (GPCRs) in peptide hormone signaling.	[K] / [CO3]
14.	List the various hormones produced by the gastrointestinal system.	[K] / [CO3]
15.	Why do heavy metals (e.g., lead, mercury) interfere with enzyme activity?	[E] / [CO5]
16.	Define Michaelis constant (Km) and its significance.	[U] / [CO5]
17.	What is the charecteristic feature of oxidoreductases	[U] / [CO5]

[3x5 = 15]

Part D

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

18.	Explain the functions and sources of Vitamin B complex vitamins.	[U] / [CO1]
19.	Describe the process of intracellular signaling involved in steroid hormone action.	[K] / [CO3]
20.	Describe the physiological significance of hormone transport mechanisms in maintaining homeostasis, considering the roles of carrier proteins and receptor types in hormone action.	[U] / [CO2]
21.	Compare and contrast the Lineweaver- Burk plot and Michaelis Menten plot with graph and explain its significance.	[E] / [CO5]
22.	Describe what happens if there is too much enzyme but not enough substrate	[U] / [CO5]
23.	Analyze the MM equation and explain the importance and siginificance of Km value	[U] / [CO5]

[5x4 = 20]